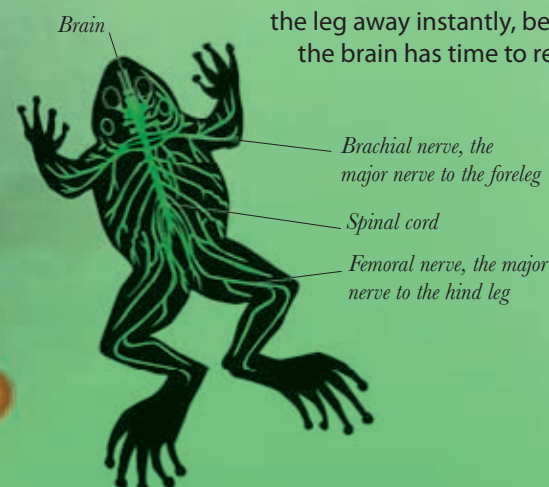




Central control

The red-eyed tree frog is only 2½ in (6 cm) long, but it has the same kind of nervous system as an elephant or a whale. These animals—and all other vertebrates—have a brain linked to a nerve highway called the spinal cord. The spinal cord connects with major nerves, which fan out to reach all parts of the body. It also controls many reflex actions, which happen without consulting the brain. One of the fastest is the withdrawal reflex. If the frog feels something painful with its leg, the withdrawal reflex makes it pull the leg away instantly, before the brain has time to react.



Giant nerves

When danger strikes, many animals make a sudden dash for safety. In cuttlefish and squid, these emergency escapes are triggered by giant nerve cells, which act much more quickly than normal-sized ones. Their fibers, or axons, are up to 0.04 in (1 mm) across, which is hundreds of times thicker than those in mammals. Earthworms also use giant nerves to escape if they are pecked by birds.

Chains of brains

A cockroach's brain contains about 1 million nerve cells and is only just big enough to cover a period. But in addition to this, it has a series of "mini-brains," or ganglia, which control the different segments of its body. The segments can act independently and can continue working for some time even if the animal loses its head. The nervous systems of other segmented animals are built on the same plan.

Main nerve cords running along the underside, beneath the digestive system

One of three thoracic ganglia, each controlling one pair of legs

Brain processes information from eyes, antennae, and other sense organs



Abdomen made up of six main segments, each controlled by a ganglion

